

14.

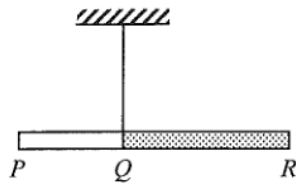


A semi-circular cardboard hangs from a spring balance from point O as shown. The reading of the spring balance is 5 N. Which statements are correct ?

- (1) The weight of the cardboard is 5 N.
- (2) The centre of gravity of the cardboard is directly under O .
- (3) The reading of the balance would become zero if the set-up is brought to the Moon's surface.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

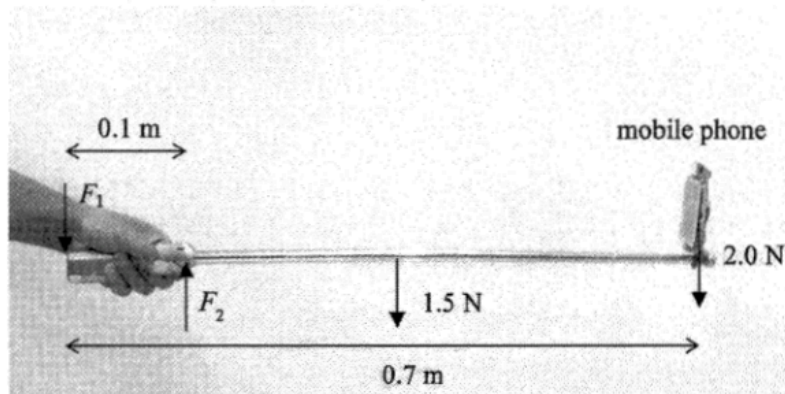
3.



PQR is a composite rod having a uniform cross-section but with portions PQ and QR made of different materials, each of uniform density. The ratio of the length of PQ to that of QR is $2 : 3$. When the rod is suspended at Q , it remains horizontal as shown. What is the ratio of the mass of PQ to that of QR ?

- A. $2 : 3$
- B. $1 : 1$
- C. $3 : 2$
- D. Answer cannot be found as the ratio of their densities is not given.

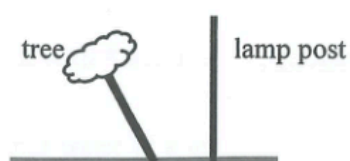
8. Selfie sticks are popular nowadays. A uniform selfie stick of length 0.7 m is held horizontally as shown. Assume that the forces required to hold the selfie stick by the hand are represented by F_1 and F_2 , and F_1 and F_2 are perpendicular to the stick.



It is given that the weight of the selfie stick and the mobile phone are 1.5 N and 2.0 N respectively. Taking the mobile phone as a point mass, estimate the magnitude of F_2 .

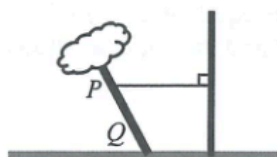
- A. 3.5 N
- B. 19.3 N
- C. 35 N
- D. Cannot be determined as F_1 is unknown.

7.

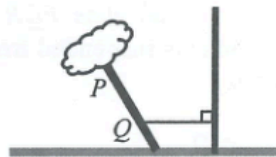


A tree blown by strong wind is found leaning to one side. In order to give the tree some support, a rope is wrapped around its trunk and tied to a fixed lamp post nearby. In which of the following arrangements would the rope have the highest chance of breaking ?

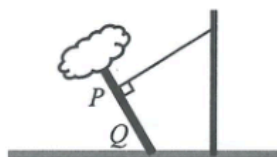
A.



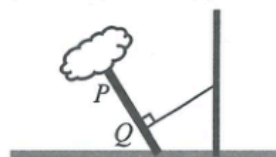
B.



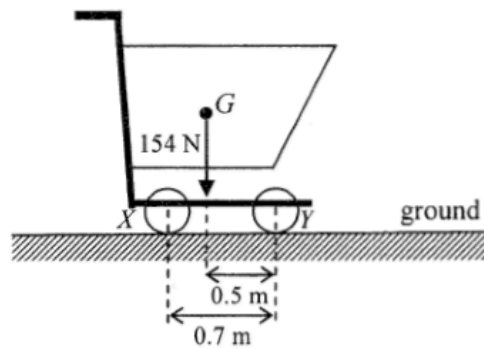
C.



D.



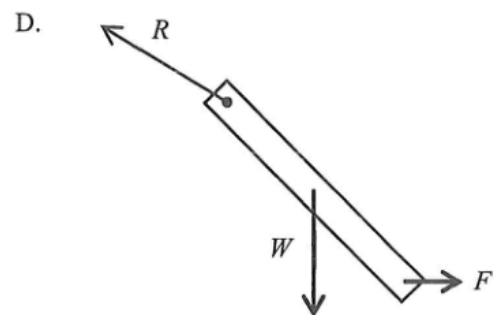
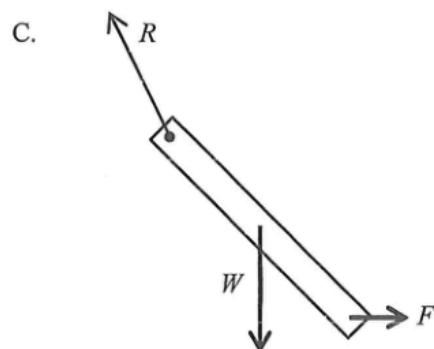
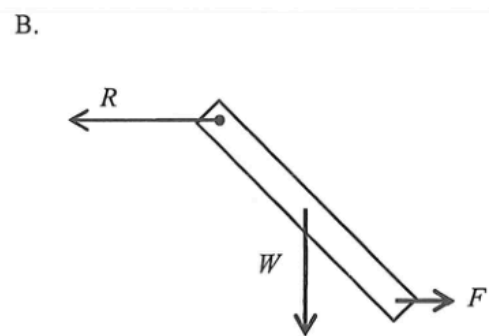
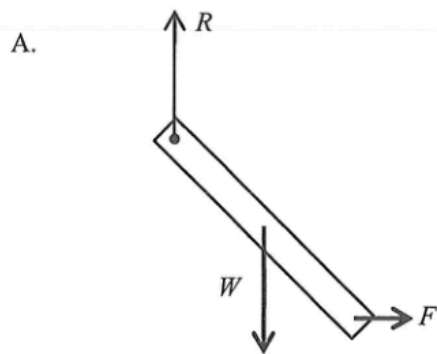
9.



The figure shows a supermarket trolley resting on the ground. The separation between cylindrical wheels X and Y is 0.7 m. When the trolley is loaded to a total weight of 154 N, its centre of gravity G is at a horizontal distance of 0.5 m from the wheel Y . What is the reaction acting on the wheel X from the ground ?

- A. 44 N
- B. 62 N
- C. 92 N
- D. 110 N

7. A uniform rod of weight W hinged at its upper end is at rest under a horizontal force F applied at its lower end. Which figure correctly shows the force R acting on the rod at the hinge ?



10. Loosening a screw requires a pair of forces of 10 N acting tangentially as shown in Figure (1). A suitable screwdriver with a handle of diameter 0.025 m shown in Figure (2) is chosen. Estimate the magnitude of the pair of forces needed to act tangentially on the handle of the screwdriver.

Figure (1)

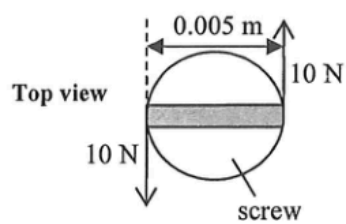
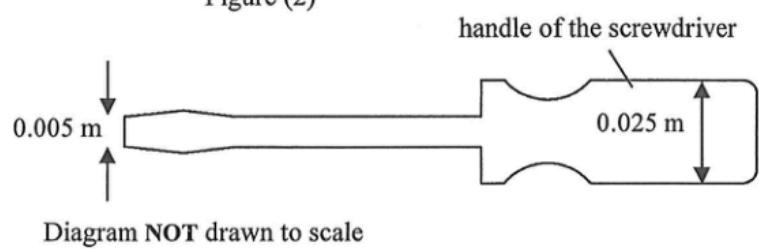
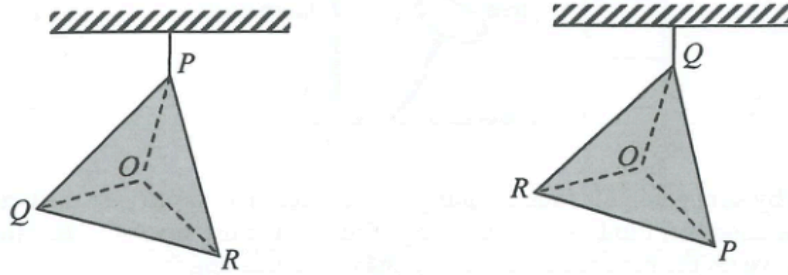


Figure (2)



- A. 2 N
- B. 4 N
- C. 5 N
- D. 10 N

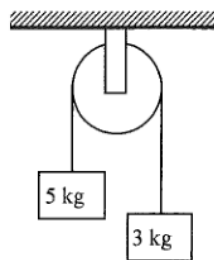
4.



O is the centre of a metal plate PQR in the form of an equilateral triangle with **non-uniform** mass distribution. The plate is suspended from the ceiling at P and then at Q as shown. The centre of gravity of the metal plate is

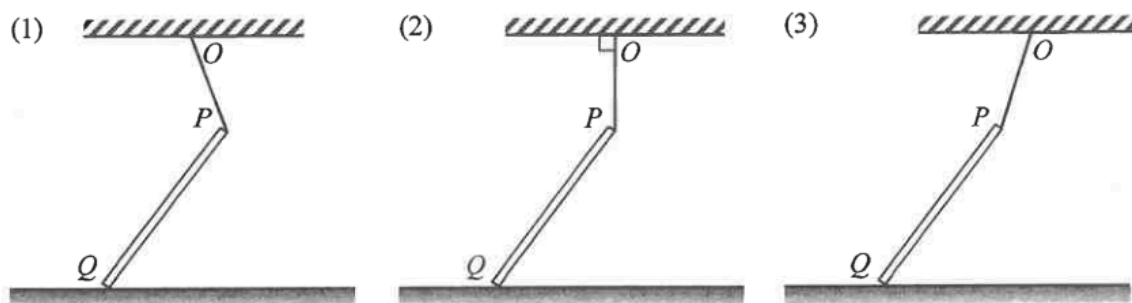
- A. at O .
- B. within the region POQ .
- C. within the region ROQ .
- D. within the region POR .

8. Two blocks of masses 5 kg and 3 kg respectively are connected by a light string passing through a frictionless fixed light pulley. Find the magnitude of the acceleration of the blocks in terms of the acceleration due to gravity g when they are released. Neglect air resistance.



- A. g
B. $\frac{g}{2}$

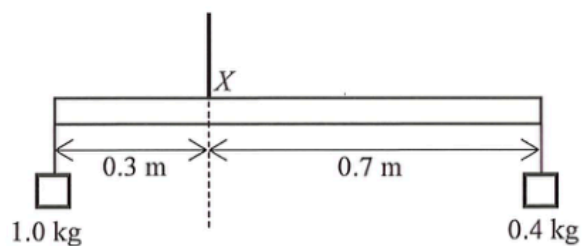
5. In each figure below, a uniform rod PQ has its upper end P attached to point O on the ceiling by a light inextensible string. The rod's lower end Q rests on a rough horizontal ground as shown.



In which figure is the frictional force acting on the rod due to the ground towards the right?

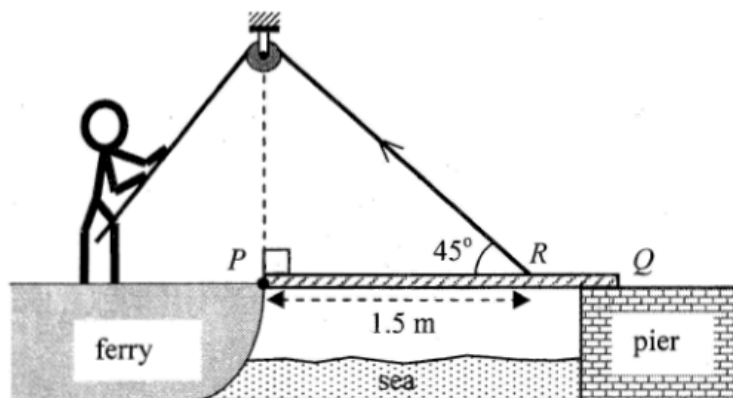
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

12. A uniform metre rule of a certain weight is hung by a string at X such that it is kept in equilibrium with weights hanging at both ends as shown. If the 1.0-kg weight is shifted 0.1 m towards X , what distance should the 0.4-kg weight be shifted towards X in order to restore equilibrium ?



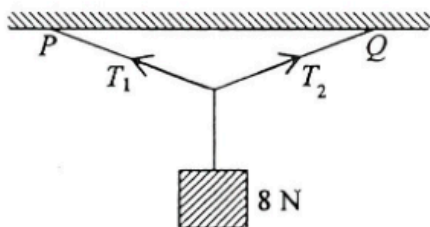
- A. 0.1 m
- B. 0.2 m
- C. 0.25 m
- D. 0.45 m

6. A uniform gangplank PQ of a ferry smoothly hinged at end P initially rests horizontally on the pier. The gangplank has mass M and length 2 m. It is raised by a man on the ferry using a light rope passing a smooth fixed light pulley and connecting to R on the gangplank as shown. R is 1.5 m from end P . Which of the following correctly describes the force required to raise the gangplank steadily?



	<i>initial</i> force required to raise the gangplank when it is horizontal	<i>subsequent</i> force required to raise the gangplank
A.	$0.67 Mg$	greater than $0.67 Mg$
B.	$0.67 Mg$	smaller than $0.67 Mg$
C.	$0.94 Mg$	greater than $0.94 Mg$
D.	$0.94 Mg$	smaller than $0.94 Mg$

5. A block of weight 8 N is suspended from a horizontal ceiling by light inextensible strings to two different points P and Q as shown. The strings are equal in length.

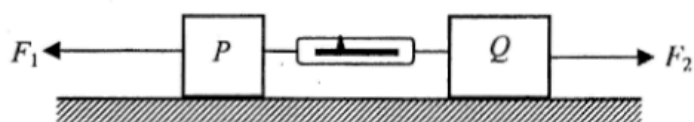


Which of the following descriptions about the tensions T_1 and T_2 in the two strings is/are correct ?

- (1) The magnitude of T_1 must be greater than 4 N.
- (2) The maximum value of T_2 would not exceed 8 N.
- (3) The resultant force of T_1 and T_2 is zero.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

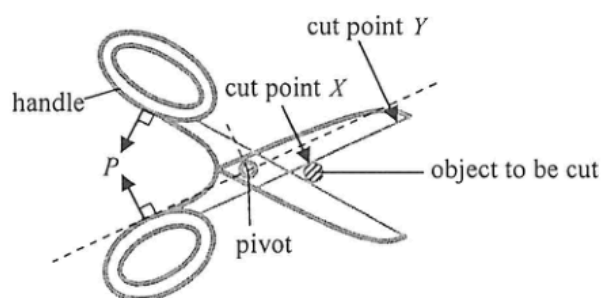
8.



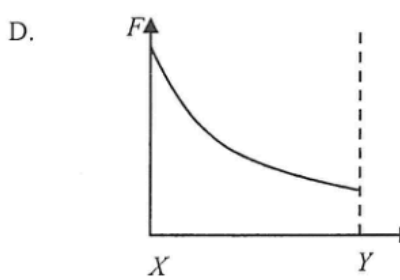
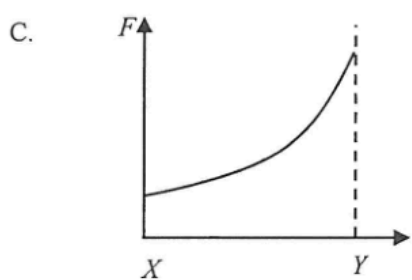
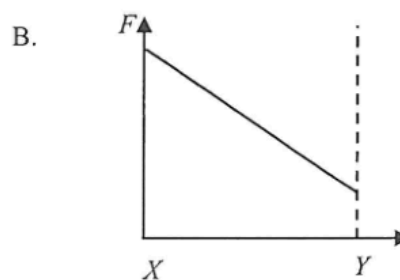
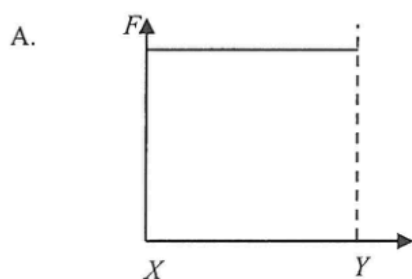
Blocks P and Q of mass m and $2m$ respectively are connected by a light spring balance and placed on a smooth horizontal surface as shown. If horizontal forces F_1 and F_2 (with $F_1 > F_2$) act on P and Q respectively and the whole system moves to the left with constant acceleration, what is the reading of the spring balance ?

- A. $\frac{2F_1 - F_2}{3}$
- B. $\frac{2(F_1 - F_2)}{3}$
- C. $\frac{2F_1 + F_2}{3}$
- D. $\frac{F_1 + 2F_2}{3}$

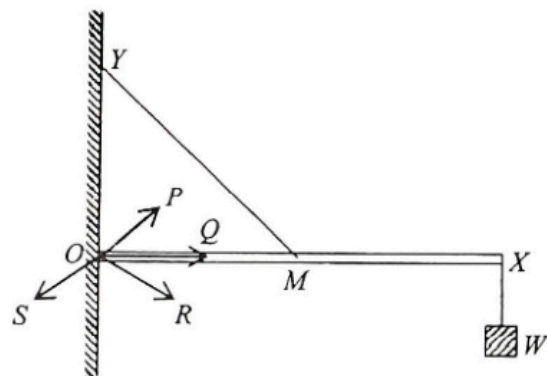
7. The figure shows that a pair of forces P of constant magnitude is applied at right angles to the handles of a pair of scissors in order to cut an object.



Which graph below best shows the variation of force F produced at cut point from X to Y when the pair of scissors is closed?



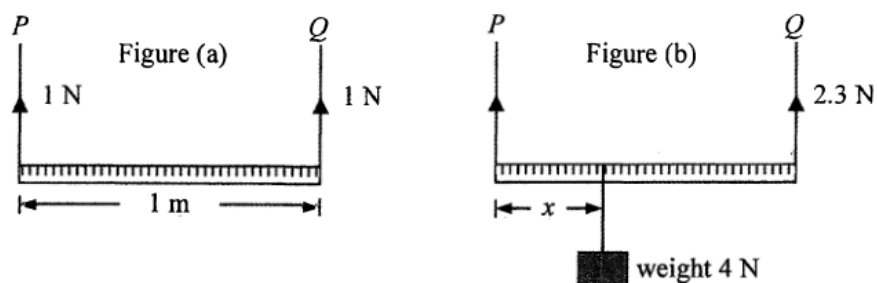
6.



A uniform light rigid rod OX is hinged smoothly to a wall at one end O . Its mid-point M is connected by a light inextensible string to a point Y directly above O while a weight W is suspended from the other end X of the rod as shown. Rod OX remains horizontal. The reaction force acting on the rod due to the wall is along the direction

- A. OP .
- B. OQ .
- C. OR .
- D. OS .

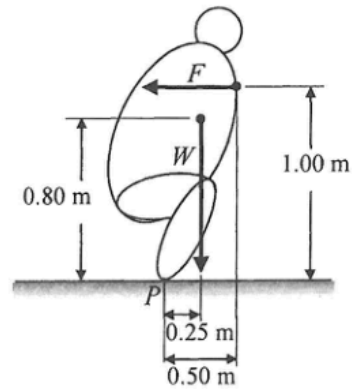
11.



A uniform metre rule is supported by vertical wires P and Q and remains at rest horizontally as shown in Figure (a). The tension in each wire is 1 N. When a weight of 4 N is hung from the metre rule at a certain position as shown in Figure (b), the tension in Q becomes 2.3 N while the metre rule remains horizontal. Find the distance x shown.

- A. 32.5 cm
- B. 57.5 cm
- C. 67.5 cm
- D. Answer cannot be found as the tension in P is not known.

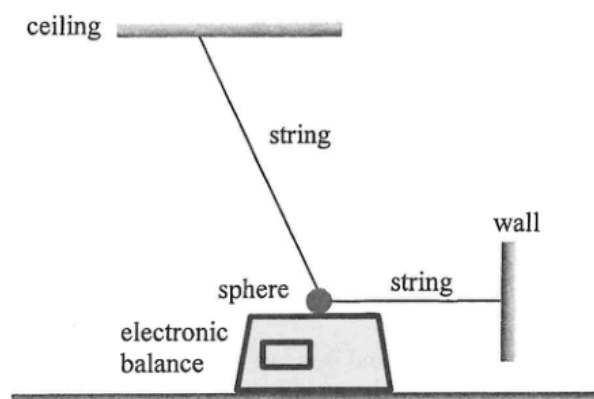
10. A child tries to push a 150 kg (sumo) wrestler in order to topple him backwards in a game. The simplified diagram indicates the weight W of the wrestler and the horizontal pushing force F acting on the wrestler by the child. P is the wrestler's contact point on the ground.



Find the minimum value of F required.

- A. 368 N
- B. 736 N
- C. 1177 N
- D. 2354 N

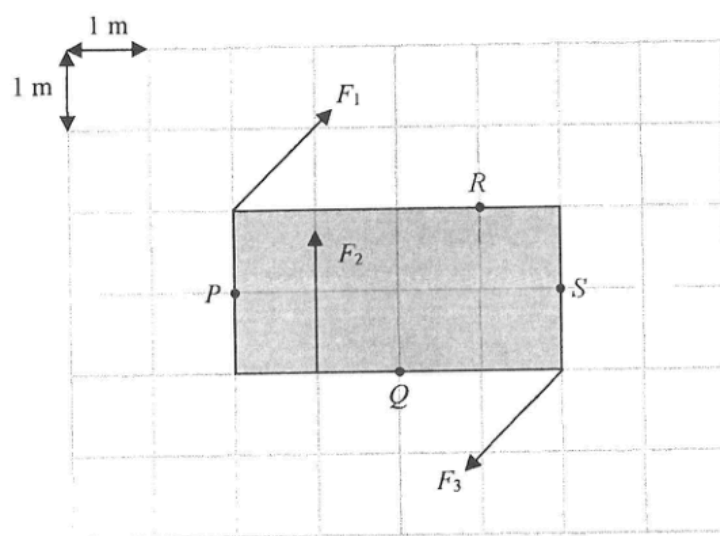
4.



A sphere of weight 18 N is suspended by two light and inextensible strings. One string is attached horizontally to a wall, and the other is connected to the ceiling. The sphere rests on the smooth top surface of an electronic balance, which shows a reading of 10 N . Find the resultant force acting on the sphere by the two strings.

- A. 0 N
- B. 8 N
- C. 10 N
- D. 15 N

7.

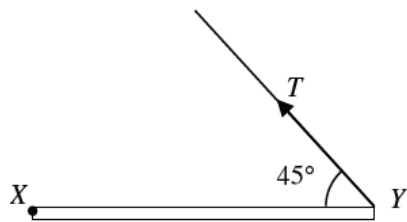


A rectangular plate of width 2 m and length 4 m is placed on a smooth horizontal surface. Three forces F_1 , F_2 and F_3 of equal magnitude act on it as shown. F_1 and F_3 are parallel and opposite in direction. Taking the moment about which point below is the net moment the greatest?

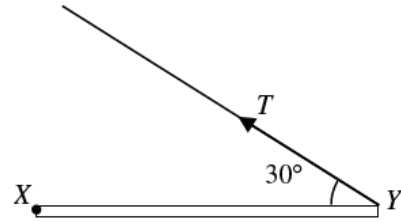
- A. P
- B. Q
- C. R
- D. S

12. A rod XY hinged at X is kept horizontal by a light string. M is the midpoint of XY . In which of the following arrangements will the tension T in the string be the smallest ?

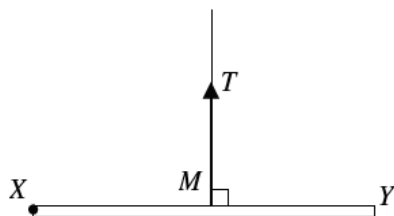
A.



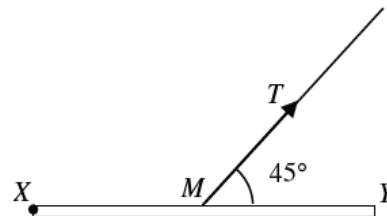
B.



C.



D.



10.

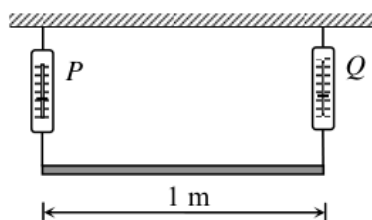


Figure (a)

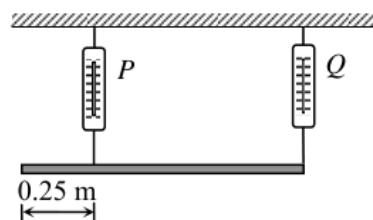


Figure (b)

Figure (a) shows a uniform plank supported by two spring balances P and Q . The readings of the two balances are both 150 N. P is now moved 0.25 m towards Q (see Figure (b)). Find the new readings of P and Q .

	Reading of P /N	Reading of Q /N
A.	100	200
B.	150	150
C.	200	100
D.	200	150